

Assignment 2: Queuing Theory

Advanced Netowrking

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1. Maximum Rate Possible

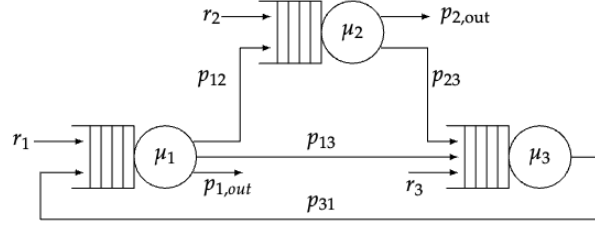


Figura 1: Queuing network diagram

Our goal is to find the maximum value for r_1 that maintains the stability of the system, that is $\lambda_i \leq \mu_i \forall i \in \{1, 2, 3\}$.

We define each arrival λ_i rate as:

$$\lambda_i = r_i + \sum_{j=1}^3 p_{ji} \lambda_j \quad (1)$$

We are given the following information:

- The arrival rate from outside of the system for servers 2 and 3: $r_2 = r_3 = 1$.
- Each routing probability p_{ij} .
- The processing rate of each server $\mu_i = 10 \text{ jobs/sec}$

Given this information we can find the maximum rate r_1 . We can model this as a linear programming problem, and since the optimum of a linear program is always achieved at a constraint boundary, we only need to check the value of r_1 when each constraint $\lambda_i \leq 10$ is saturated.

For each server i , we set $\lambda_i = 10$ and solve for r_1 , keeping only the cases where all other constraints are also satisfied. The maximum value of r_1 across these cases is our answer.

$$\begin{cases} \lambda_i \leq 10 \quad \forall i \in \{1, 2, 3\} \\ \lambda_1 = r_1 + p_{31} \lambda_3 = 1 + 0,8 \lambda_3 \\ \lambda_2 = r_2 + p_{12} \lambda_1 = 1 + 0,8 \lambda_1 \\ \lambda_3 = r_3 + p_{13} \lambda_1 + p_{23} \lambda_2 = 1 + 0,2 \lambda_1 + 0,2 \lambda_3 \end{cases} \quad (2)$$

1.1. Saturate output of server 1

$$\begin{cases} \lambda_1 = 10 \\ \lambda_2 = r_2 + p_{12} \lambda_1 = 9 \\ \lambda_3 = r_3 + p_{13} \lambda_1 + p_{23} \lambda_2 = 4,8 \\ r_1 + p_{31} \lambda_3 = 10 \Rightarrow r_1 = 4,2 \end{cases} \quad (3)$$

All constraints are satisfied and we obtain that when $\lambda_1 = 10$, $r_1 = 4,2$.

1.2. Saturate output of server 2

$$\lambda_2 = r_2 + p_{12} \lambda_1 = 10 \Rightarrow \lambda_1 = 11,25 \quad (4)$$

If we staurate λ_2 , we break the λ_1 constraint, so this is not a feasible solution.

1.3. Saturate output of server 3

$$\lambda_3 = r_3 + p_{13} \lambda_1 + p_{23} \lambda_2 = 10 \Rightarrow \lambda_1 + \lambda_2 = 45 \quad (5)$$

The sum of λ_1 and λ_2 is 45, so one of them must be breaking the constraint, making this solution unfeasible.

1.4. Optimal result

If we saturate λ_2 or λ_3 , we reach that some of the other constraints are not satisfied. Therefore, the optimal solution is obtained when saturating λ_1 , with which we obtain $r_1 = 4,2$.

2. Little's Theorem

2.1. Average occupation

We can use Little's Theorem to obtain the average number of patients in the waiting room N , we have the mean waiting time $T = 180$ minutes and the arrival rate $\lambda = 0,2$ patients/minute. So we obtain:

$$N = \lambda T = 36 \text{ patients} \quad (6)$$

2.2. Size of the waiting room

It is not possible to determine a waiting room size that will always accommodate all arrivals based solely on the given information. Patient arrivals are random, meaning there is no fixed upper bound on how many patients could arrive in a short period of time.

Therefore, in theory, an arbitrarily large number of patients could arrive, so no finite waiting room size can guarantee accommodation in all cases. However, if additional information about the arrival distribution were provided, we could design the waiting room to handle demand with a specified confidence level.

3. M/M/1 Queue Length

Here N_Q denotes the length of the queue. In the state 0, there are no packets in the system, so the queue is empty. In the state 1 too, as the arrived packet is in the server slot, so the queue is empty. For any state $i \geq 2$, the length of the queue is $i - 1$. So we can define the expectance of the queue length as:

$$E[N_Q] = \sum_{i=2}^{+\infty} (i-1) P_i \quad (7)$$

With P_i the stationary probability of state i , using the development from the notes: $P_n = \rho^n P_0$ and that $P_0 = (1 - \rho)$, we arrive at the following:

$$E[N_Q] = \sum_{i=2}^{+\infty} (i-1) \rho^i (1 - \rho) = (1 - \rho) \sum_{i=2}^{+\infty} (i-1) \rho^i = (1 - \rho) \rho \sum_{i=2}^{+\infty} (i-1) \rho^{i-1} = \quad (8)$$

$$= (1 - \rho) \rho \sum_{i=1}^{+\infty} i \rho^i = (1 - \rho) \rho \sum_{i=0}^{+\infty} i \rho^i = \frac{\rho^2}{1 - \rho} \quad (9)$$

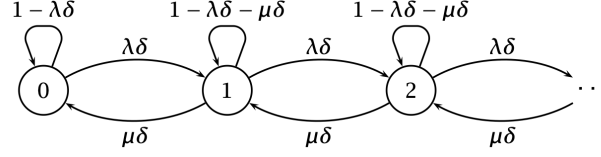


Figura 2: M/M/1 Markov chain

4. M/M/1 Queue

We have packets arriving to a link at a rate of 450 packets/sec, each having 500 bytes on average. The link capacity link is 1 Mbps.

$$1\text{Mbps} = 0,125 \cdot 10^6 \text{ bytes/sec} \Rightarrow 500\text{packets/sec} \quad (10)$$

So we have the following information:

- Arrival rate (λ): 450 packets/sec.
- Service rate (μ): 500 packets/sec.
- Number of servers (s): 1.
- Utilization rate (ρ): $\frac{\lambda}{\mu} = 0,9$.

4.1. Queue length

We need to obtain the probability that the queue has one, two or ten packets. That is that the probability that the system occupancy is two, three or eleven packets respectively (P_2 , P_3 and P_{10}). We can use the same expression from the last exercise: $P_n = \rho^n(1 - \rho)$.

- $P_2 = \rho^2(1 - \rho) = 0,081$.
- $P_3 = \rho^3(1 - \rho) = 0,0729$.
- $P_{11} = \rho^{11}(1 - \rho) \approx 0,0314$.

4.2. Mean number of packets

We can obtain the mean number of packets in the system as the expectance of the queue occupancy n :

$$E[n] = \sum_{k=0}^{+\infty} k P_k = \sum_{k=1}^{+\infty} k \rho^k (1 - \rho) = \frac{\rho}{1 - \rho} = 9 \quad (11)$$

And the mean number of packets in the queue as the expectance of N_Q :

$$E[N_Q] = \frac{\rho^2}{1 - \rho} = 8,1 \quad (12)$$

4.3. Mean waiting time

We can obtain this using Little's theorem, given mean number of packets in the queue $E[N_Q]$ and the arrival rate λ :

$$T = \frac{E[N_Q]}{\lambda} = 0,018 \text{ sec} \quad (13)$$